

Driftwood Arcade Browser Game Portal

Category: Web Development

1. Project Overview

Driftwood Arcade is a small independent studio that makes light hearted pixel art games for the web. This document describes a responsive multi page gaming website where visitors browse a catalogue of games, open a game, play a round directly in the browser, and check a high score board. The site is a static front end build that uses HTML, CSS, and plain JavaScript, with no backend, no database, and no account system. The goal of this document is to capture everything needed to build the portal and the three games it ships with.

2. Project Snapshot

Field	Value
Category	Web Development
Project type	Responsive multi page gaming website
Pages	Eight linked pages sharing one header and footer
Playable games	Three, each on an HTML5 canvas
Technology	HTML5, CSS3, vanilla JavaScript ES6
Score storage	Browser local storage
Catalogue source	assets/games.json loaded with fetch
Hosting	Any static file server
Backend	None

3. Audience and Goal

The audience is casual players who want to play in the browser with no download and no sign up. Success is a short path from the home page to a played round, a catalogue that loads from data, three working games that each save a best score, and a layout that holds from a 320 pixel phone up to a wide desktop.

4. Site Map and Pages

Page	File	Purpose
Home	index.html	Intro, featured games, link into the catalogue
Catalogue	games.html	Grid of all games from the dataset, filter and search

Dune Dash	dune-dash.html	One button endless jumper on a canvas
Star Catcher	star-catcher.html	Falling object catcher on a canvas
Tide Match	tide-match.html	Four by four memory game
High scores	leaderboard.html	Best score per game from local storage
About	about.html	Studio story and the three game descriptions
Contact	contact.html	Styled form with browser side validation

5. Visual Theme

The studio has no existing palette, so the palette below is the one to use. Buttons use the Sand background with Midnight text. Links and active states use the Tide colour. Error messages use the Coral colour.

Purpose	Name	Hex
Page background	Midnight	#161A2B
Panels and cards	Surface	#232845
Primary accent	Sand	#F4A259
Secondary accent	Tide	#4ECDC4
Primary text	Foam	#F5F6FA
Muted text	Mist	#9AA0BF
Loss and errors	Coral	#E15A5A
Win and success	Kelp	#6FCF6B

Logo and headings use a pixel style display font under the SIL Open Font License with a monospace fallback. Body text uses the system sans stack at a 16 pixel base size and a 1.5 line height. The centred content area is at most 1100 pixels wide, the header is 64 pixels tall, and the gap between cards in any grid is 20 pixels.

6. Games Catalogue and Dataset

The catalogue page is built from `assets/games.json`. The page loads the dataset with `fetch` and renders one card per game. The cards are not written into the HTML by hand.

1. Each card shows the thumbnail, the title, a category badge, the difficulty, and the short description.
2. A card with status `playable` shows a Play button that links to its game page.
3. A card with status `coming_soon` shows a muted Coming soon label and no Play button.
4. A category filter offers All, Action, Arcade, Memory, and Puzzle and shows only matching cards.

5. A text search filters cards by title while ignoring letter case, and the filter and search work together.

6. When no card matches, the page shows the line No games match that filter.

7. Game Specifications

7.1 Dune Dash

1. The canvas is 800 pixels wide by 300 pixels tall.

2. Space, the Up arrow, and a tap or click each make the character jump, and it cannot jump again until it lands.

3. Touching an obstacle ends the run.

4. The score counts up by 1 for every obstacle that passes, and the obstacle speed increases after every 5 passed.

5. On game over the canvas shows the final score, the best score, and a restart control.

6. The best score is saved under the local storage key `driftwood_dune_dash_best`, and a higher score replaces it.

7.2 Star Catcher

1. The canvas is 600 pixels wide by 400 pixels tall.

2. The basket moves with the Left and Right arrows and follows the pointer across the canvas.

3. Catching a star adds 1 to the score, and catching a bomb removes 1 life from a start of 3 lives.

4. The fall speed increases as the score rises.

5. The run ends at 0 lives, then the canvas shows the final score, the best score, and a restart control.

6. The best score is saved under the local storage key `driftwood_star_catcher_best`, and a higher score replaces it.

7.3 Tide Match

1. The board is a four by four grid of 16 cards forming 8 pairs, drawn from the supplied sprite icons.

2. Clicking a face down card turns it face up, a matched pair stays up, and a mismatch turns both down after a short pause.

3. The board shuffles at the start of every new game.

4. A moves counter increases by 1 each time a second card of a pair is turned up.
5. The game ends when all eight pairs match and shows a win message with the moves used.
6. The fewest moves are saved under the local storage key `driftwood_tide_match_best`, and a lower number replaces it.

8. Other Features

1. The leaderboard shows one row each for Dune Dash, Star Catcher, and Tide Match, with the label Best score for the first two and Fewest moves for the third, and a dash when a value is missing.
2. The contact form has Name, Email, and Message fields, all required, with an email pattern check and a minimum of 10 characters in the message.
3. On submit the contact page prevents the default submit, shows inline errors in Coral for invalid fields, and on a valid submit shows the line Thanks for reaching out to Driftwood Arcade.
4. The layout shows no horizontal scrollbar from 320 pixels wide upward, the header links collapse behind a menu button at 600 pixels or narrower, and the catalogue grid moves from one to two to three columns as width grows.
5. Each game canvas scales down to fit its container width on small screens while keeping its width to height ratio.

9. Deliverables

1. Eight HTML pages as listed in section 4, sharing one header and one footer.
2. One or more CSS files for the theme, with no inline style attributes used for layout.
3. JavaScript files for the catalogue, the three games, the leaderboard, and the form.
4. The dataset kept in place at `assets/games.json`.
5. The supplied art kept in place under `assets/sprites`.
6. A README file that explains how to run the site on a local static server.

10. Folder Structure

The delivered site uses the folder layout below. The exact split of JavaScript files is open as long as each page loads only the scripts it needs and the catalogue reads the dataset from `assets/games.json`.

```
driftwood-arcade/  
  index.html  
  games.html  
  dune-dash.html  
  star-catcher.html  
  tide-match.html  
  leaderboard.html  
  about.html  
  contact.html  
  css/  
    style.css  
  js/  
    main.js  
    catalogue.js  
    dune-dash.js  
    star-catcher.js  
    tide-match.js  
    leaderboard.js  
    contact.js  
  assets/  
    games.json  
    sprites/  
  README.md
```

11. Assets Provided

File	What is in it
reference_image.png	Layout mockup of the home page and the catalogue
assets/games.json	Games dataset with six records, three playable and three coming soon
assets/sprites/logo.png	Driftwood Arcade logo for reference
assets/sprites/logo-mark.png	Square logo emblem for a favicon or badge
assets/sprites/thumb-*.png	Six catalogue card thumbnails, one per game
assets/sprites/runner.png, obstacle.png	Dune Dash character and obstacle sprites
assets/sprites/basket.png, star.png, bomb.png	Star Catcher basket, star, and bomb sprites
assets/sprites/card-back.png	Tide Match card back sprite
assets/sprites/icon-*.png	Eight memory icons for the Tide Match pairs
instructions.md	The full written brief with every rule

Assets: https://drive.google.com/drive/folders/1-gU5PnqMlc-kg_NYPdOCNeevYDizq0i?usp=sharing

12. Acceptance Criteria

1. All eight pages are present and link to each other through the shared header.
2. The catalogue renders its cards from assets/games.json using fetch.
3. The category filter and the title search both work on the catalogue.
4. Dune Dash runs, ends on collision, and stores its best score under driftwood_dune_dash_best.
5. Star Catcher runs, ends at 0 lives, and stores its best score under driftwood_star_catcher_best.
6. Tide Match ends when all pairs match and stores the fewest moves under driftwood_tide_match_best.
7. The leaderboard shows the three stored values and a dash when a value is missing.
8. The contact form validates its three fields and shows the confirmation line on a valid submit.
9. The layout shows no horizontal scrollbar from 320 pixels upward and the header collapses at 600 pixels or narrower.
10. The browser console shows no errors during normal use.
11. The README explains how to run the site on a local static server.

13. Out of Scope and Revisions

1. No backend, database, or login system.
2. No multiplayer, online leaderboards, or chat.
3. No payment, advertising, or purchase features.
4. No framework, bundler, or paid plugin, and no paid or branded art or fonts.

One round of revisions is included. All feedback is collected into a single message after the first build is shared. Any change outside the rules in this document is not part of this project. The final handover is the complete source folder of HTML, CSS, JavaScript, the dataset, the assets, and the README, delivered as one archive that runs when served from a static file server.

End of document.